



AI at the Edge: **Semantic Search** **and Device Memory** **with Qdrant**

GenAI Zurich - Chadha Sridi - April 2026

Why Edge AI?

Three Real-World Constraints

Unreliable Connectivity



Pharma Cleanrooms

Real-Time Reaction



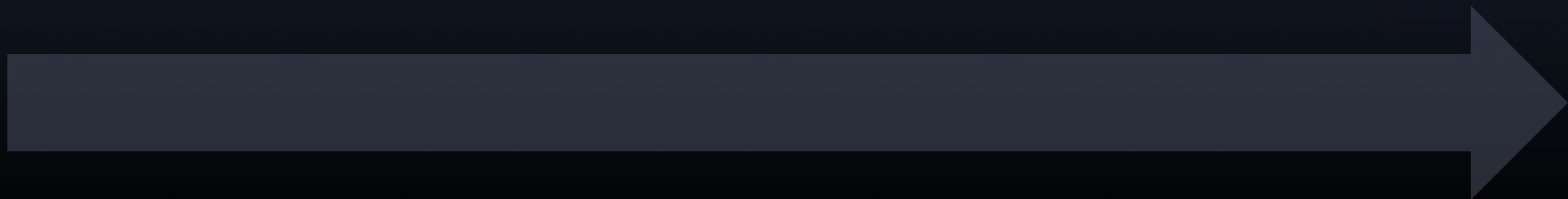
Autonomous Vehicles

Data Privacy



Personal Devices

Edge AI Maturity



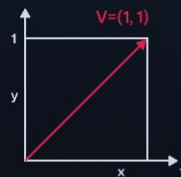
Vector Search

Two different vector embeddings should be close to each other if they represent a similar input object.

Embeddings are generated by neural networks and can represent thousands of dimensions.

Embeddings can represent any kind of data: text, images, video, code, audio, etc.

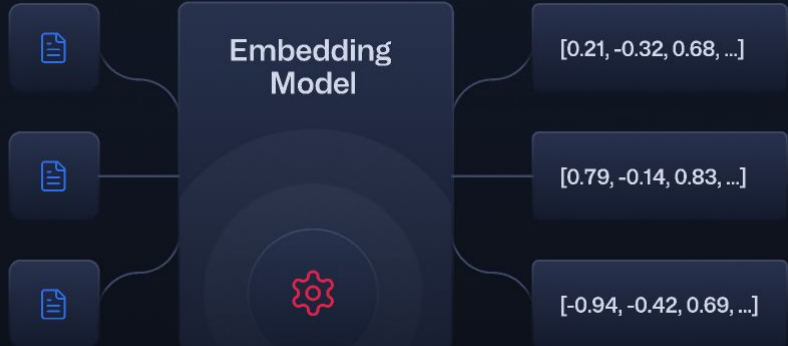
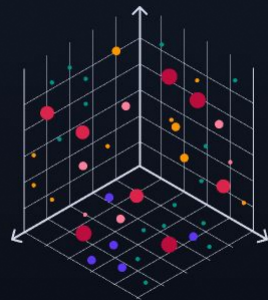
Qdrant handles embeddings natively.



2-Dimensional
Vector



3-Dimensional
Vector



Data Objects

Vector Embeddings

Qdrant Edge

Bring vector search to the edge: an embeddable, high-performance engine that runs directly on mobile and other edge devices.

- Run on **low-CPU** devices.
- Use one API to manage and synchronize data **on-device** and in your **cloud cluster**.
- Fit common on-device cases: **phones** and **laptops**, smart-home/**IoT**, **robotics**.

Key features:

- Use **local storage** to avoid network latency.
- Support **multi-tenant** setups; treat each device as its own tenant.
- Embed as a library; runs in-process with no background **daemons**.



Qdrant Edge Anomaly Detection

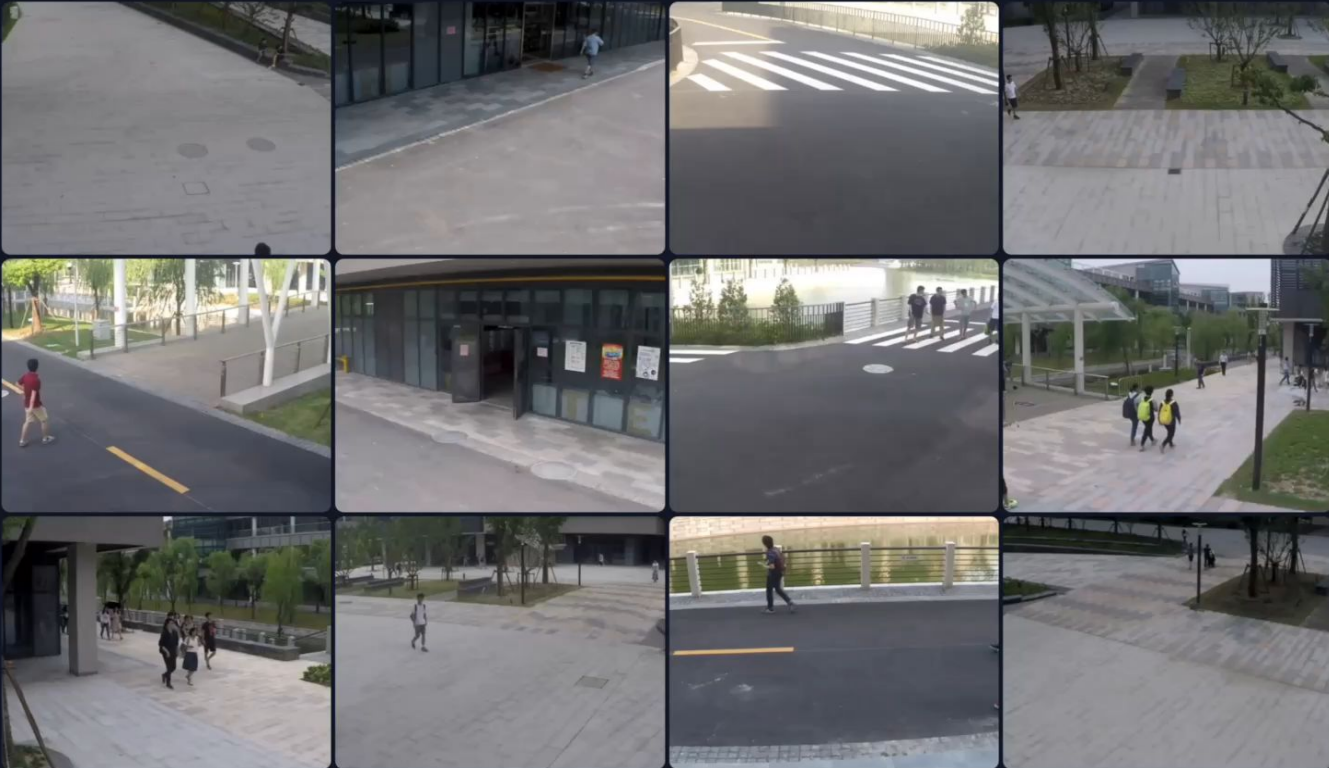
**Qdrant Edge**

Sentinel
Video anomaly detection
Twelve Labs + Qdrant

[Cameras](#) [Incidents](#) [Search](#) [Chat](#) [Intelligence](#) ⓘ

[Take a tour](#)



Campus Alerts 61 new

12 cameras · 3 zones · 61 alerts

☐ Hide reviewed

1 **99**

Five people engage in physical fight
Main Entrance · Zone A
Fighting 2:12:04 PM · 20s
Paved forecourt, multiple approach lines

2 **98**

Two individuals fight on crosswalk
Library Plaza · Zone A
Fighting 2:11:36 PM · 13s
Roadway with zebra crossing

3 **97**

Group fight erupts while two people ru...
North Path · Zone B
Fighting 2:12:21 PM · 24s
Curving roadway with crosswalk

4 **94**

Multiple people cross the pedestrian a...
Library Plaza · Zone A
Running 2:11:48 PM · 15s
Roadway with zebra crossing

5 **94**

Four people run quickly across crossw...
Library Plaza · Zone A
Running 2:13:32 PM · 10s
Roadway with zebra crossing

6 **89**

Ten people walk along sidewalk, includ...
East Gate · Zone A
Vehicle 2:11:31 PM · 14s
Glass façade, sidewalk and curb lane

Man exits with phone, cyclist passes b...

POWERED BY

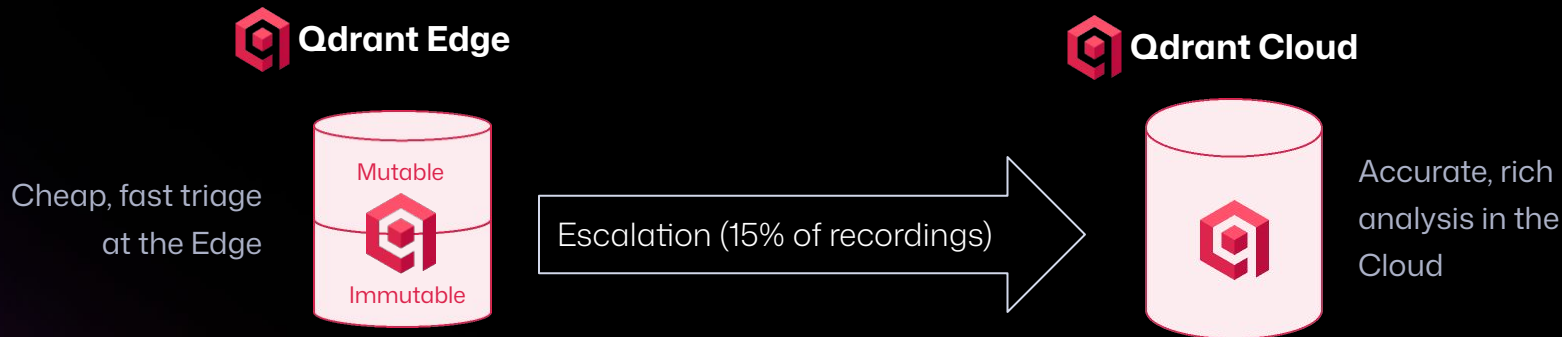
 Qdrant Edge

 Twelve Labs

 Vultr

 NVIDIA

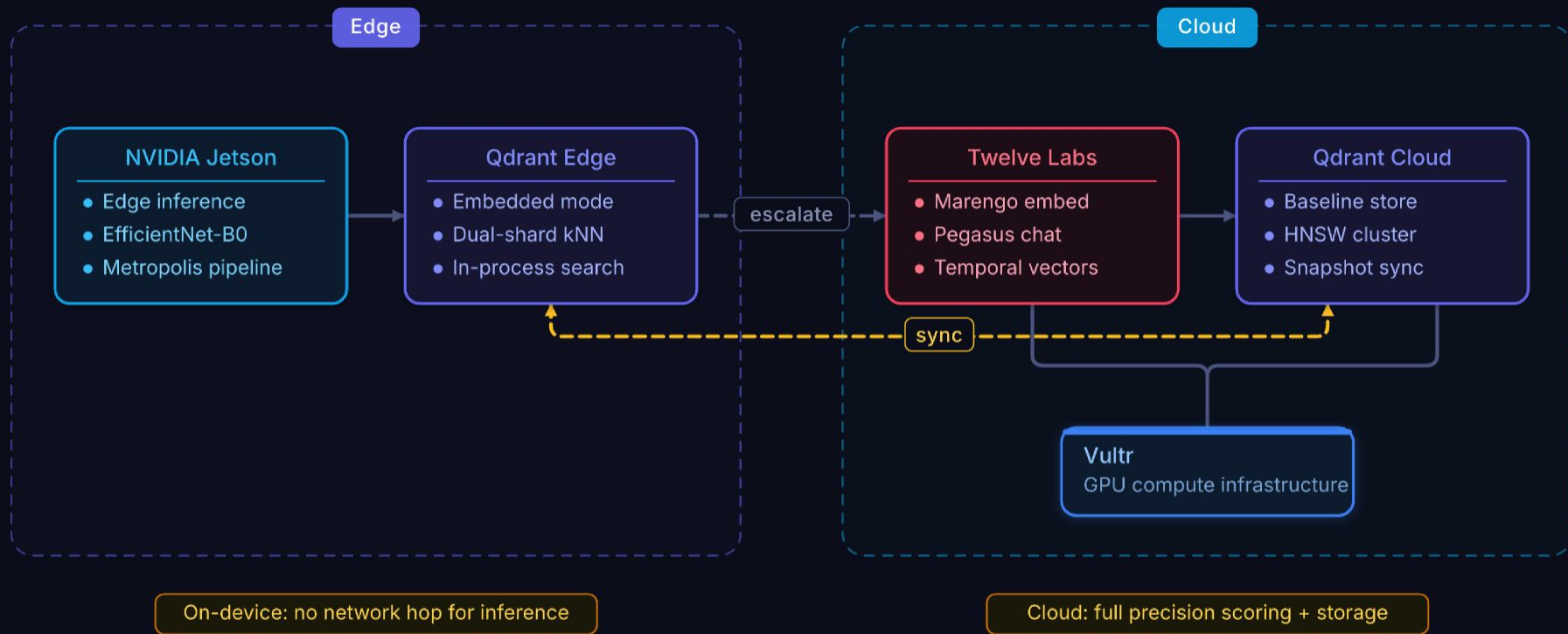
How it Works



Why kNN?

- Open world coverage
- No model training
- Concept drift

How it Works



□ Inference □ Vector search □ Video intelligence □ Compute

Benefits

- Local first
- Lower latency
- Lower cloud compute
- Lower cloud storage
- Lower costs

~6x less cloud processing volume with ~95% accuracy



Thank You!



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